NARRATIVE QUARTERLY REPORT
(APRIL–JUNE) 2026

Project — In situ monitoring of water quality in Cuban bays: creating and promoting the use of scientific tools

Participating institutions

- University of Havana (UH)
- Université Libre de Bruxelles (Belgium)

Cooperating institutions

- Center for Environmental Research and Transport Management (CIMAB)
- ESPOLETA Tecnologías (SME)

Other collaborating institutions

- José Antonio Echeverría Technological University of Havana (Cujae)
- Havana Bay State Management Group
- Université Catholique de Louvain

Project coordinators
Cuban side: Dr. C. Ernesto Altshuler Álvarez
Foreign side: Dr. Denis Terwagne

Participants — Dr. Geydis Fundora Nevot, Dr. Florence Degavre, Dr. Stamatios Nicolis, Dr. Alexandre Campo, Lic. Daniela Martínez Ortiz (PhD candidate), Eng. Alex Rivera Rivera (PhD candidate), MSc. Luis Abel Rodríguez de Torner (PhD candidate), Dr. Aramis Rivera Denis, Eng. Orestes Chávez Linares, Dr. Dayaris Hernández.

Duration — Five years

General objective — To provide the Cuban scientific-social ecosystem with a set of tools that enables in situ monitoring of water-quality parameters in Cuban bays, in a sustainable way and connected to local communities.

Specific objectives

- Design and build a wave channel at the Faculty of Physics, UH
- Design and build a portable toolkit for in situ measurement of Havana Bay's water quality
- Set up a server with a website to receive data from the buoys and from the community
- Establish a methodology for using the buoys
- Train the corresponding human resources

Fulfillment of the quarterly objectives

Task 1. Detecting and addressing vulnerabilities in the Physics building that endanger the project

Efforts have continued to eliminate the vulnerabilities of the Physics building on the University of Havana's hill campus, where multiple windows in very poor condition due to termite damage make it easy for potential intruders to get in. This is a VERY SERIOUS situation. Given the lack of response through institutional channels, on June 23 we proceeded, using our own resources, to board up another window in very poor condition facing the front of the building — one of the windows of the room where the ARES project's wave channel is to be built.

Trying to reduce the vulnerability of the rooms associated with the ARES project. The photo was taken on June 23, 2026, after reinforcing, with our own efforts, the window on the left, which was about to collapse toward the front of the building (Photo: E. Altshuler).

Task 2. Drafting the purchase list, and managing goods acquired with project funding

The first container from Brussels arrived at the port of Mariel (Cuba) in April 2026. Its contents are to be installed in the rooms mentioned under Task 1, per the project's plan. Given the country's fuel shortage, the importing company (EMIDICT) has not yet collected the container; we met with company officials on June 30 to try to speed up the process. We continue monitoring the process for the second container from Belgium and the second shipment from Panama. It should also be noted that the company ALIANC Group Servicios Integrales (https://www.aliancgroup.com), and in particular the specialist Leonardo Durán, still owes the project the software for the spectrophotometer acquired with project funds; according to this person, the software has been "stuck" at DHL Cuba for several weeks, though the problem has not been properly tackled.

Task 3. Continued instrumentation for in situ measurements

PhD candidates Alex Rivera and Daniela Martínez, with the participation of Dr. Aramis Rivera and Dr. Dayaris Hernández, made notable progress on several key fronts, detailed below.

In situ spectrophotometry. Calibration of the low-cost, open-access, in-house spectrophotometer was completed. PhD candidates Daniela Martínez and Alex Manuel Rivera worked closely together on this front, using the professional spectrophotometer acquired with project funding as the reference instrument for calibration. The process consisted of calibrating the designed spectrophotometer using colored solutions of known concentrations; the signals obtained from the "low-cost" spectrophotometer were systematically compared against measurements from the commercial spectrophotometer for the same solutions, a methodology that allows sufficiently precise calibration curves to be established. The preliminary results obtained form the experimental basis for a new scientific article now in preparation.

Bio-fouling contamination studies. As part of the studies on biofouling and its impact on optical measurement of water-quality parameters, an experiment was designed and run to assess contamination of protected versus unprotected glass surfaces using copper mesh. The experiment consisted of preparing six paired sample holders: in each pair, one unprotected glass and one glass protected with a copper mesh, to assess the mesh's effectiveness as a protective barrier against biofouling and the resulting effect on optical transmittance. On May 15, 2026, all sample holders were placed at the same depth in the bay, specifically at the Geocuba pier. The experimental protocol called for collecting one sample holder every 2–3 days to monitor the evolution of surface contamination; during the exposure period, 2 collections were carried out following the established contamination-assessment protocol, with samples analyzed on the spectrophotometer to quantify the effect on transmittance as a function of exposure time and accumulated biofouling. However, on the seventh day of the experiment, an unexpected phenomenon was detected that led to suspending the trial: the copper mesh underwent a chemical reaction, becoming coated in copper chloride, giving it a characteristic blue-green color and forming crystals of this salt on its surface. This process compromised the mesh's integrity and, therefore, the validity of the experiment in its original design. Following these results, the remaining samples were withdrawn from the bay and a new approach for this study is being worked on.

Detail of the corrosion effect on the copper mesh exposed to seawater. Copper chloride formation (blue-green coloring) is observed after 7 days of exposure. Each paired sample holder has a control and a sample (a slide without and with copper-mesh protection, respectively).

Shadowmetry + machine learning on liquid-contaminant slicks on water. 4th-year student Jean Rafael López, supervised by E. Altshuler, ran experiments on the surface dispersion of chemical contaminants — specifically fuel oil — using the "shadowmetry" method aided by machine learning, to determine contaminant-slick shape under laboratory conditions. The left column shows experimental images of several slicks obtained directly via shadowmetry; the right column shows the same images automatically segmented after feeding them to an artificial-intelligence algorithm. The work won First Prize in the Electronics and Computing section of the 2026 edition of the Faculty of Physics' Student Scientific Conference (UH), as well as a special mention across the whole conference.

Writing scientific articles on in situ monitoring. In April, the review article titled "Microorganisms as indicators of marine pollution: challenges for in situ monitoring" was submitted for consideration to the journal Environmental Monitoring and Assessment. During April, May, and June, PhD candidate Daniela Martínez has been handling the revisions requested by the journal's editor; this manuscript, which comprehensively summarizes the state of the art in detecting microorganisms as bioindicators of marine pollution and the technological challenges of in situ monitoring, has been under peer review since June 14, awaiting the journal's final decision. Work also continued on a second scientific article comparing low-cost, open-access spectrophotometer designs; started in previous months, it underwent a rewriting process in June to better target its intended audience, per the supervisors' recommendations, and is currently in its first rounds of supervisor review. Once the suggestions are incorporated, the article is expected to be submitted for publication next quarter.

Task 4. Preliminary experiments on the dynamics of solid contaminants in a wave channel

PhD candidate Luis Abel Rodríguez continued his stay at ULB, where he continued the work on floating contaminants under wave action carried out in Cuba and reported in the January–March report, in the wave channel he built at the Belgian laboratory. In particular, the following actions were carried out.

Wave-channel characterization. Characterization of the FabLab-ULB wave channel was completed as a function of the period T (0.40–0.60 s) and piston stroke S (10–45 mm), using a lateral optical setup that extracts the free-surface elevation $\eta(x, t)$ from each video. A fixed side camera (1920×1080 px, 30 fps) records a region of interest on the water surface; a vision algorithm detects the free-surface line frame by frame and locates its crests and troughs, allowing wave height H and effective period to be calculated automatically. From the 31 (T, S) combinations tested, the corresponding wave height H and wavelength λ were obtained, placing the resulting regimes on the Le Méhauté diagram to classify the type of wave generated under each condition. Three wave-absorber geometries (perforated flat, parabolic, and interchangeable plates with different porosity) were also designed and tested to minimize reflection at the channel end opposite the piston — a necessary step before extending the characterization sweep to the reflection coefficient Cr.

Floating-cylinder experiments. Experiments were run with series of balanced and unbalanced floating cylinders (series M and M1, mean mass 29.85 ± 0.36 g and 29.47 ± 0.11 g respectively, with at-rest imbalance angles between 0.3° and 17°), under wave conditions with H between 6 and 11 mm. Each run was recorded with a side camera and the cylinder's position tracked via a color-based (HSV) detection pipeline, generating a (t, x, y) time series; detection gaps (195 frames, 0.18% of the total) were corrected manually via a graphical interface. From these trajectories, transit time and mean drift velocity Vm = d/t along the channel were calculated. Results show that the at-rest imbalance angle measurably affects both quantities: the more unbalanced cylinders (series M1) systematically drift faster than the balanced ones (series M), consistent with the wave-driven orientation and drift mechanisms reported in the literature — matching the experimental results obtained with real empty cans in our lab's wave channel in Havana, reported in the previous quarterly report.

Wave-driven drift of a scale cylinder. Top: 3D-printed cylinders with different flotation angles. Bottom: cylinder floating in the wave channel, with its trajectory in red — note the Stokes drift toward the "beach" (on the right).

Numerical simulations. 3D simulations were developed in DualSPHysics (SPH, ~930,000 particles: 726,000 fluid and 203,000 boundary, of which 228 form the floating cylinder), run on ULB/CECI's Lyra GPU cluster (NVIDIA RTX 6000 Ada, 48 GB). Piston motion was imposed from the calibrated motion file corresponding to T = 0.6 s, S = 45 mm, with the cylinder left free only to rotate about the channel's longitudinal axis, locked in all other directions. Two cases were simulated, balanced (m = 10.29 g) and unbalanced (m = 11.71 g, tilt 5.47°), for Tmax = 30 s. Both qualitatively reproduce the pitching-and-drift behavior observed experimentally under the same piston-generated wave train, closing the characterization → physical experiment → numerical replica loop. As shown in the simulations, unbalanced cylinders advance toward the beach faster than balanced ones, matching the experiments carried out in the Cuban wave channel with real empty cans, reported in the January–March 2026 report.

Wave-driven drift of scale cylinders: DualSPHysics simulations. Top: balanced cylinder. Bottom: unbalanced cylinder. In each image, the cylinder's initial position is shown on the left, and its position after a given time interval on the right.

Next steps: complete characterization of the reflection coefficient Cr across the full (T, S) sweep; extend numerical simulations to all 31 experimental conditions and compare them quantitatively against the experimental trajectories; carry out a convergence study with finer particle resolution; and advance the design and construction of the new FabLab FF-UH wave channel, together with designing new experiments with floating and non-floating objects, and liquid contaminants.

Task 5. Academic exercises carried out by the PhD candidates involved in the Project

PhD candidate Daniela Martínez-Ortiz successfully completed her categorization exercise as an instructor professor in May, consolidating her teaching career within the Faculty of Physics. PhD candidate Luis Abel Rodríguez de Torner gave a PhD progress seminar at ULB on June 26, 2026. The academic exercises carried out by the social-sciences PhD candidate are included in the next section.

Task 6. Continued research in the social sciences

Continuation of Mayelín García Remus's doctoral stay at the Université Catholique de Louvain-la-Neuve (February–August 2026). She took part in seminars held at CIRTES (the Interdisciplinary Research Center on Work Organization and Society), where she is doing her placement, on topics including: social innovation and exnovation; a discussion of an article on women entrepreneurs and the feminist movement; social-inclusion policies; the discomfort qualitative research can cause the researcher during fieldwork; perceived inequalities in business opportunities and strategies; and governance, learning, and temporality. She also took part in a CIRTES-organized workshop on vulnerabilities and gender-based violence. Progress continued on the doctoral thesis's theoretical chapter: continuing the literature review on social innovation, water-quality monitoring technologies, water governance, participatory monitoring, and citizen science, searching for scientific articles in Google Scholar, Scielo, and Science Direct based on defined inclusion/exclusion criteria to identify the articles most relevant to the research. In detail: (a) selecting and analyzing articles to inform the theoretical framework and the article to be written, using the PRISMA method, a methodological contribution to the thesis; (b) defining search strings ("social innovation" AND "water pollution" AND "community participation"), ("social innovation" AND "water quality monitoring"), ("technologies" AND "water quality monitoring"), ("social innovation"); (c) identifying roughly 330 articles per search string meeting the defined inclusion criteria — for example, 144 peer-reviewed articles in English and Spanish on social innovation, water pollution, and community participation, published between 2016 and 2026; using the PRISMA checklist, 15 were then selected from each string.

Conclusion. Despite the country's dramatic economic situation, which has seriously eroded the participants' living conditions, cut off all access to existing buoys, and left infrastructure with zero maintenance from outside the team's efforts, the project is reaching such a level of maturity that a growing ramp of scientific results is expected in the coming months.

Sincerely,
Dr. C. Ernesto Altshuler
Project Director, Cuban side.

Machine-assisted English translation of the original Spanish narrative report submitted to ARES. Figures and photographs are reproduced in the original Spanish document.